

Problem 18.9

A level perpetuity-immediate is to be shared by A , B , C , and D . A receives the first n payments, B the second n payments, C the third n payments, and D all payments thereafter. It is known that the ratio of the present value of C 's share to A 's share is 0.49. Find the ratio of the present value of B 's share to D 's share.

Solution

Let the level payment be 1, and let $v = (1 + i)^{-1}$. Then the present values of the shares are

$$A = a_{\overline{n}|i},$$

$$B = a_{\overline{n}|i}v^n,$$

$$C = a_{\overline{n}|i}v^{2n},$$

and

$$D = a_{\infty|i}v^{3n}.$$

We are given that

$$\frac{C}{A} = 0.49.$$

Since

$$\frac{C}{A} = \frac{a_{\overline{n}|i}v^{2n}}{a_{\overline{n}|i}} = v^{2n},$$

we have

$$v^{2n} = 0.49.$$

Thus,

$$v^n = 0.7.$$

Now calculate the desired ratio:

$$\frac{B}{D} = \frac{a_{\overline{n}|i}v^n}{a_{\infty|i}v^{3n}}.$$

Using

$$a_{\overline{n}|i} = \frac{1 - v^n}{i} \quad \text{and} \quad a_{\infty|i} = \frac{1}{i},$$

we get

$$\begin{aligned} \frac{B}{D} &= \frac{\left(\frac{1 - v^n}{i}\right)v^n}{\left(\frac{1}{i}\right)v^{3n}} \\ &= \frac{1 - v^n}{v^{2n}}. \end{aligned}$$

Substituting $v^n = 0.7$ and $v^{2n} = 0.49$,

$$\begin{aligned}\frac{B}{D} &= \frac{1 - 0.7}{0.49} \\ &= \frac{0.30}{0.49} \\ &= \frac{30}{49}.\end{aligned}$$

Hence, the ratio of the present value of B 's share to D 's share is

$$\boxed{\frac{30}{49}}.$$